Blepharitis



Lorne Yudcovitch, OD, MS, FAAO

SOAP Format

- S = Subjective
  - AKA symptoms
  - Everything the patient says
- O = Objective
  - AKA signs
  - All the doctors observes and all the test results
- A = Assessment
  - AKA diagnosis (or diagnoses)
  - What is wrong with the patient
- P = Plan
  - For each diagnosis, what is going to be done to help the patient

Blepharitis

- **Anterior blepharitis - inflammation centered around eyelashes and follicles**
  - **Staphylococcal** – lives on the skin (Grahms positive org.)
  - **Seborrheic**
  - **Mixed**
- **Posterior blepharitis - inflammation involving meibomian glands and gland orifices**
  - **Meibomian seborrheic**
  - **Meibomian gland dysfunction (AKA meibomianitis, meibomitis)**
- **Angular blepharitis**
  - **Staphylococcal**
  - **Moraxella**

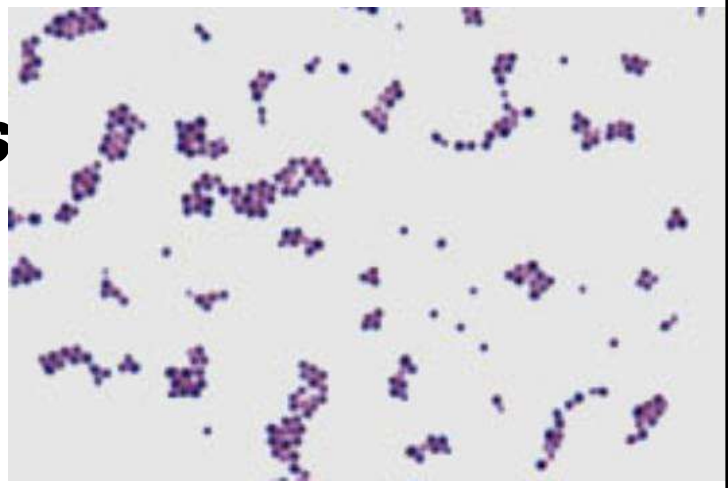


Staphylococcal Blepharitis

Etiology

Infection of the lash follicles and lid margins

- **With *Staphylococcus aureus* and/or**
  - ***S. aureus* is normally present on lid margins of small number of individuals**
- **overgrowth of *Staphylococcus epidermidis***
  - ***S. epidermidis* normally present on lid margins (normal flora) of most individuals**
- **→ inflammatory response by the body to this bacteria**

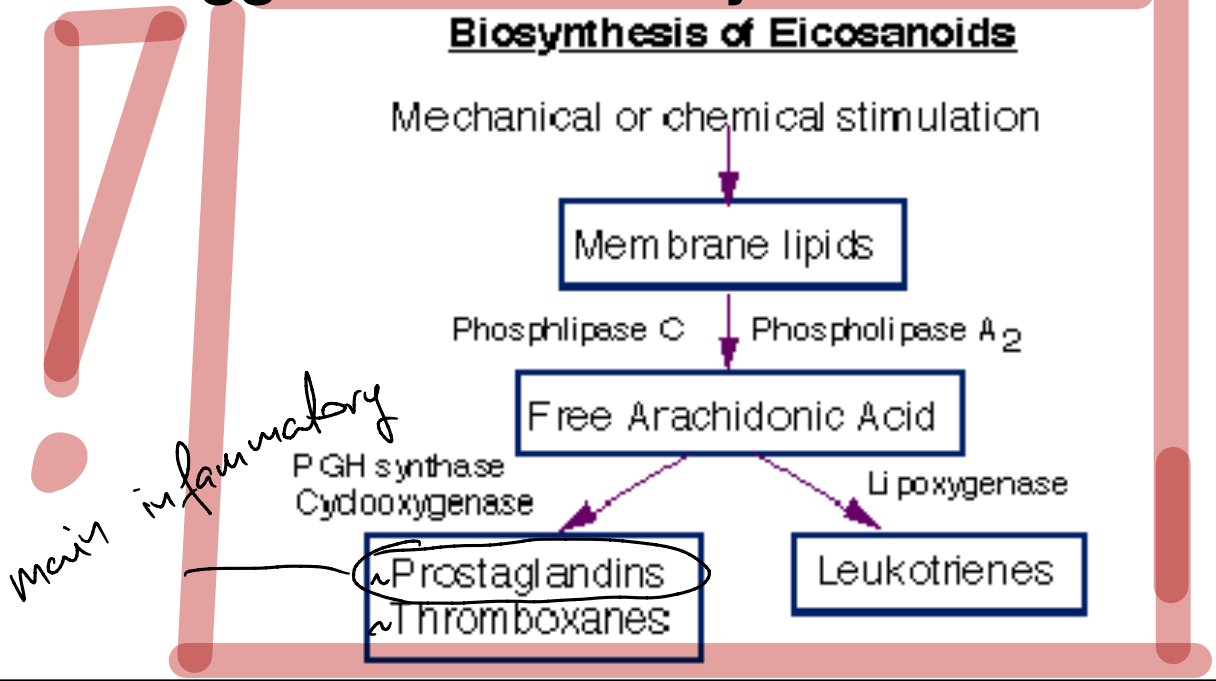


GRAHM POSITIVE

Staphylococcal Blepharitis

nuclear = white

- **Secondary mechanism of inflammation**
  - **Production by the bacteria of lipase**
  - **→ breaks down the lipids in the tears**
  - **→ free fatty acids (FFAs)** – lipid layer
  - **→ trigger inflammatory and irritative response**



main inflammatory

inflammatory response

Staphylococcal Blepharitis

Inflammatory response:

- **Dilation of blood vessels (redness)**
- **Increased vascular permeability (edema + blood proteins)**
- **Movement of PMNs from blood into tissue**
- **If inflammatory response is caused by bacteria, lots of PMNs move from blood into tissues to fight infection through phagocytosis**
- **→ dead PMNs + protein clot = pus**



- **<u>**** Presence of pus = bacterial infection</u>**

PMNs = polymorphonucleocytes (white blood cells)

Staphylococcal Blepharitis

Subjective

- Depends on severity
- Chronic irritation and burning
- FB (foreign body sensation)
- Tender lids (may have recurrent hordeola or chalazia)
- Patient may notice crusts on lashes
- The lids may be stuck together in morning
- Patient may complain of red, irritated lids

Ocular Adnexa

Images used: 2.2, 2.5, 2.10, 2.1, 2.8, 2.9, 2.12, 2.14

I. External features of the eyelids

A. Palpebral fissure

Thin

B. Lateral and medial canthus

1. Plica semilunaris

2. Caruncle

C. Eyelid Margin

1. Cilia

2. Pores of the meibomian glands

II. Anatomy of the eyelids

A. Skin

B. Palpebral conjunctiva

1. Fornix

2. Goblet cells

C. Mucocutaneous junction

D. Tarsal plate

1. Dense connective tissue that gives the eyelid rigidity and structure

2. The superior tarsal plate is approximately 8 mm high in Asians and 10 mm in Caucasians.

3. The inferior tarsal plate is approximately 5 mm high

4. Clinical correlate: Eyelid eversion

E. Orbital septum

1. The orbital septum is a sheet of connective tissue that starts at the orbit rim and inserts into the levator aponeurosis above the tarsal plate superiorly and the tarsal plate inferiorly.

2. Function

    a. Separates the eyelid structures from the orbital contents.

    b. Prevents eyelid infections from entering the orbit and therefore having a direct connection to the brain.

F. Eyelid muscles

1. Orbicularis oculi muscle

    a. Encircles the palpebral fissure.

    b. Functions

        (1) Closes the eyelids

        (2) Protects the eye

        (3) Renews the tear film

        (4) Keeps the eyelid margin near the globe

        (5) Helps with meibomian gland secretion

    c. Clinical correlate: ectropion and entropion

    d. Innervation: cranial nerve 7

2. Superior palpebral levator muscle

    a. The levator muscle starts in the posterior orbit.

    b. It travels forward with the superior rectus muscle.

    c. Levator aponeurosis

        (1) Tendinous expansion of the levator

    d. Function

        (1) Elevation of the eyelids

        (2) Connections between the superior rectus and the levator results in increased eyelid height when looking up.

    e. Innervation: cranial nerve 3

3. Müller muscle

    a. Smooth muscle

    b. Inserts into the tarsal plate

    c. Function

        (1) Additional eyelid elevation

    d. Innervation: sympathetic nervous system

4. Clinical correlate: Ptosis
    a. A drooping of the upper eyelid.

G. Glands of the eyelids

1. Meibomian glands

    a. Embedded in the tarsal plate

    b. Function: secrete the outer lipid layer of the tear film

2. Zeis glands

    a. Secretes sebum into the hair follicle

    b. Function: keep the eyelash from becoming brittle.

3. Accessory lacrimal glands

    a. Located in the upper and lower conjunctival fornix and orbital borders of the tarsal plate

    b. Function: contributes to the aqueous layer of the tear film

III. Lacrimal system

A. Tear film

1. Functions

a. Lubricant between the globe and eyelids

b. Helps remove debris

c. Source of oxygen

d. Antibacterial substances

e. Maintains corneal hydration

f. Regulate ocular surface wound repair

2. Layers

a. Lipid layer

(1) Produced in meibomian glands

(2) Function

i. Retards evaporation

ii. Provides lubrication

iii. Stabilizes the tear film (lowering surface tension) to keep tears from overflowing down the cheeks

b. Aqueous layer

(1) Secreted by the lacrimal and accessory lacrimal glands

c. Mucin layer

(1) The surface of the conjunctival and corneal cells has microvilli covered with this glycocalyx layer which is hydrophilic.

(2) Function

i. Facilitates adhesion of the aqueous layer to the ocular surface

ii. Allows the aqueous layer to spread evenly over the surface of the eye

B. Nasolacrimal drainage system

1. Puncta

2. Canaliculi

3. Lacrimal sac

4. Nasolacrimal duct

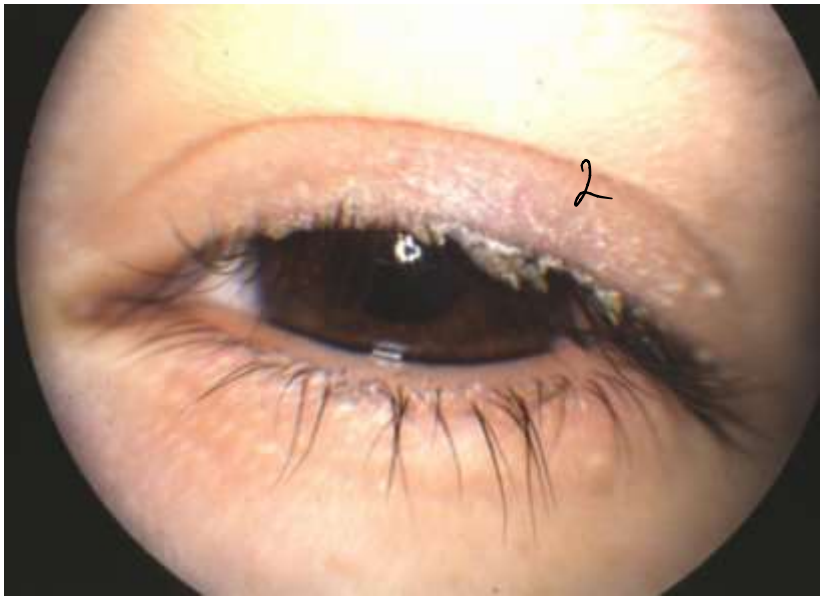
5. Valve of Hasner: found at the end of the nasolacrimal duct.

Staphylococcal Blepharitis

Objective

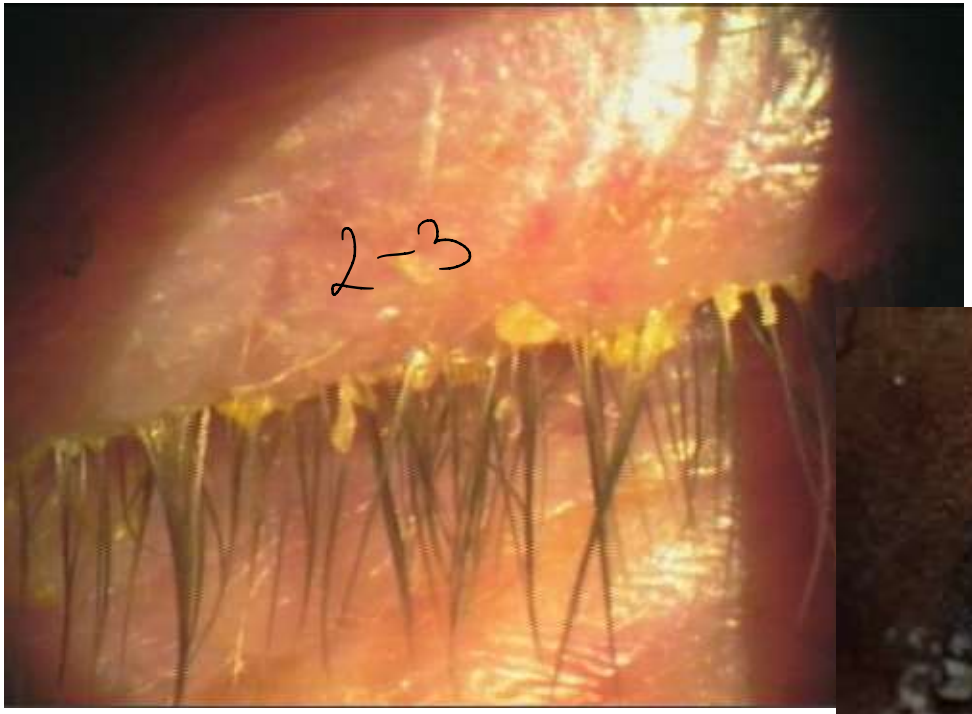
- **Crusts along the lash line at the base of the lashes**
- **Dried pus**





Staphylococcal Blepharitis

More crusts





Staphylococcal Blepharitis

2-

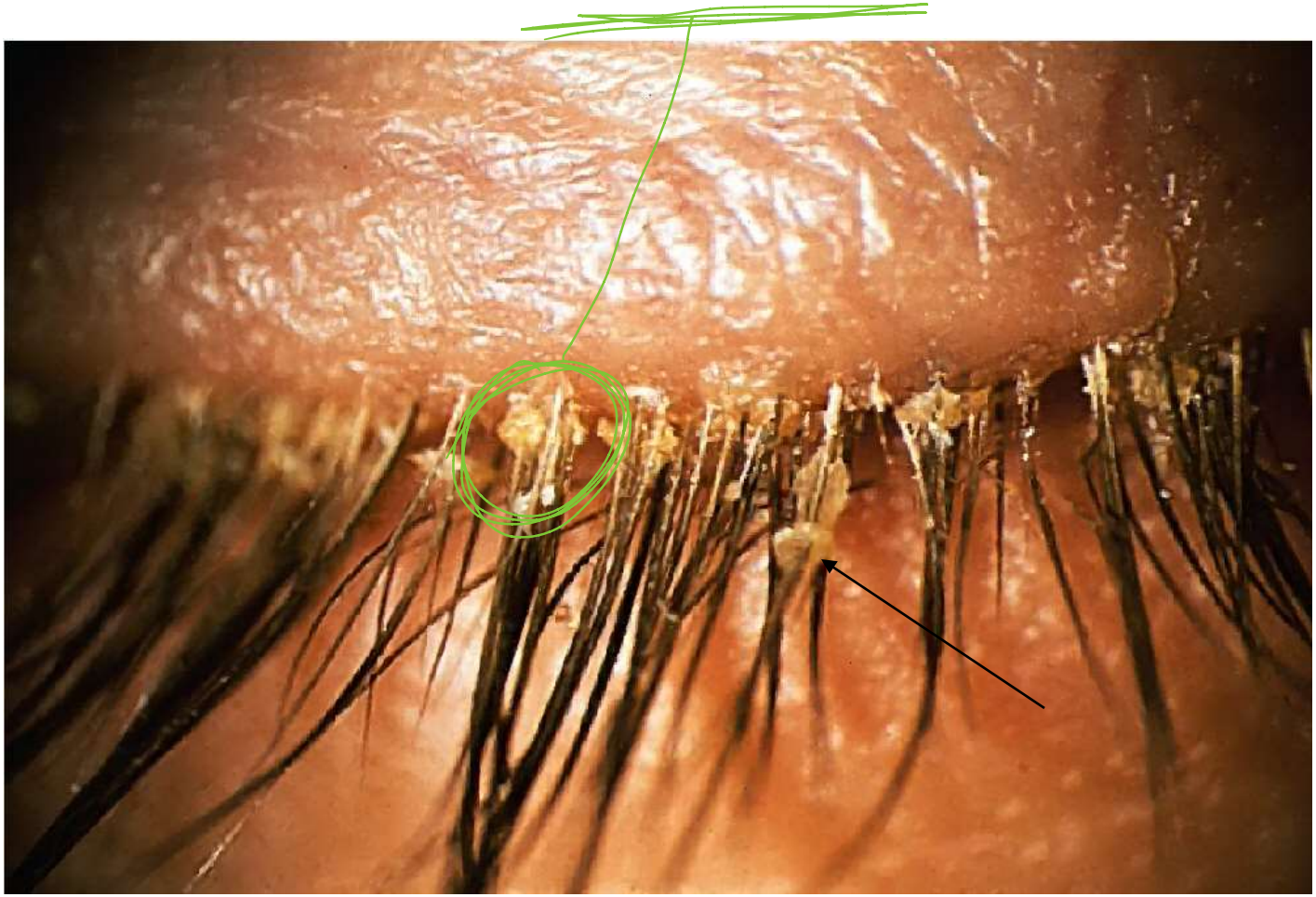


4+



Staphylococcal Blepharitis

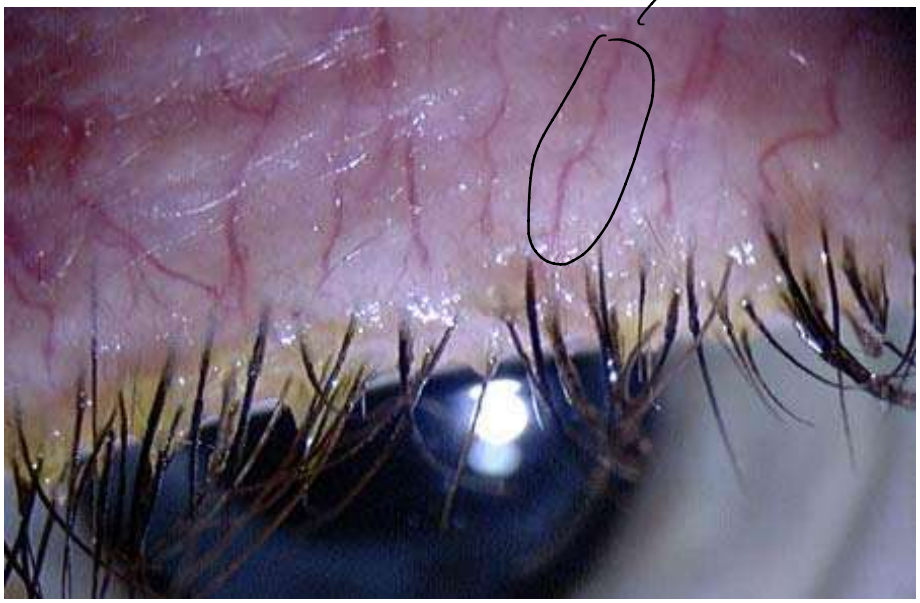
- Crust can encircle lash = collarette



Staphylococcal Blepharitis

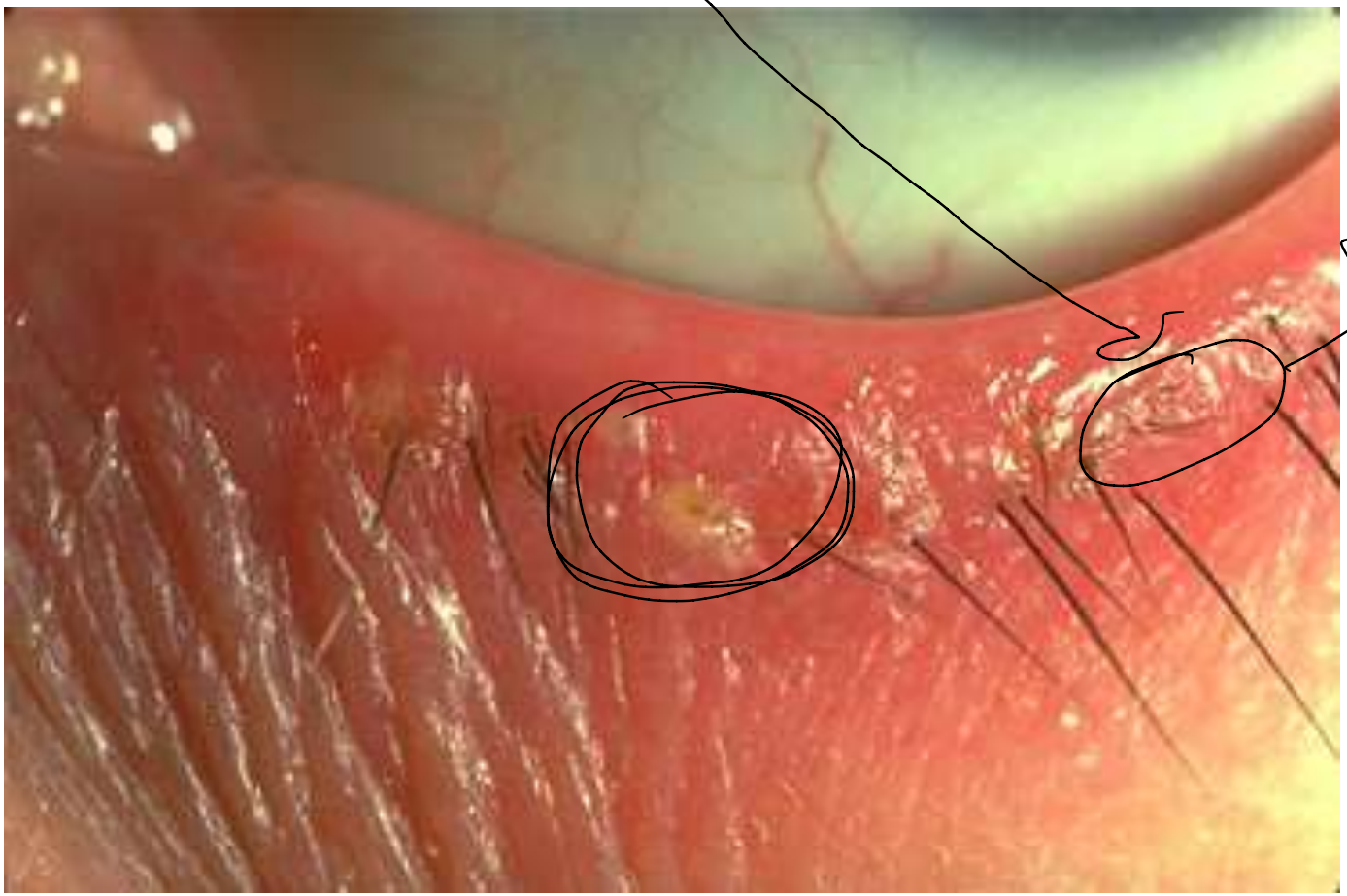
- Crust at the lash base – can also appear ‘oily’





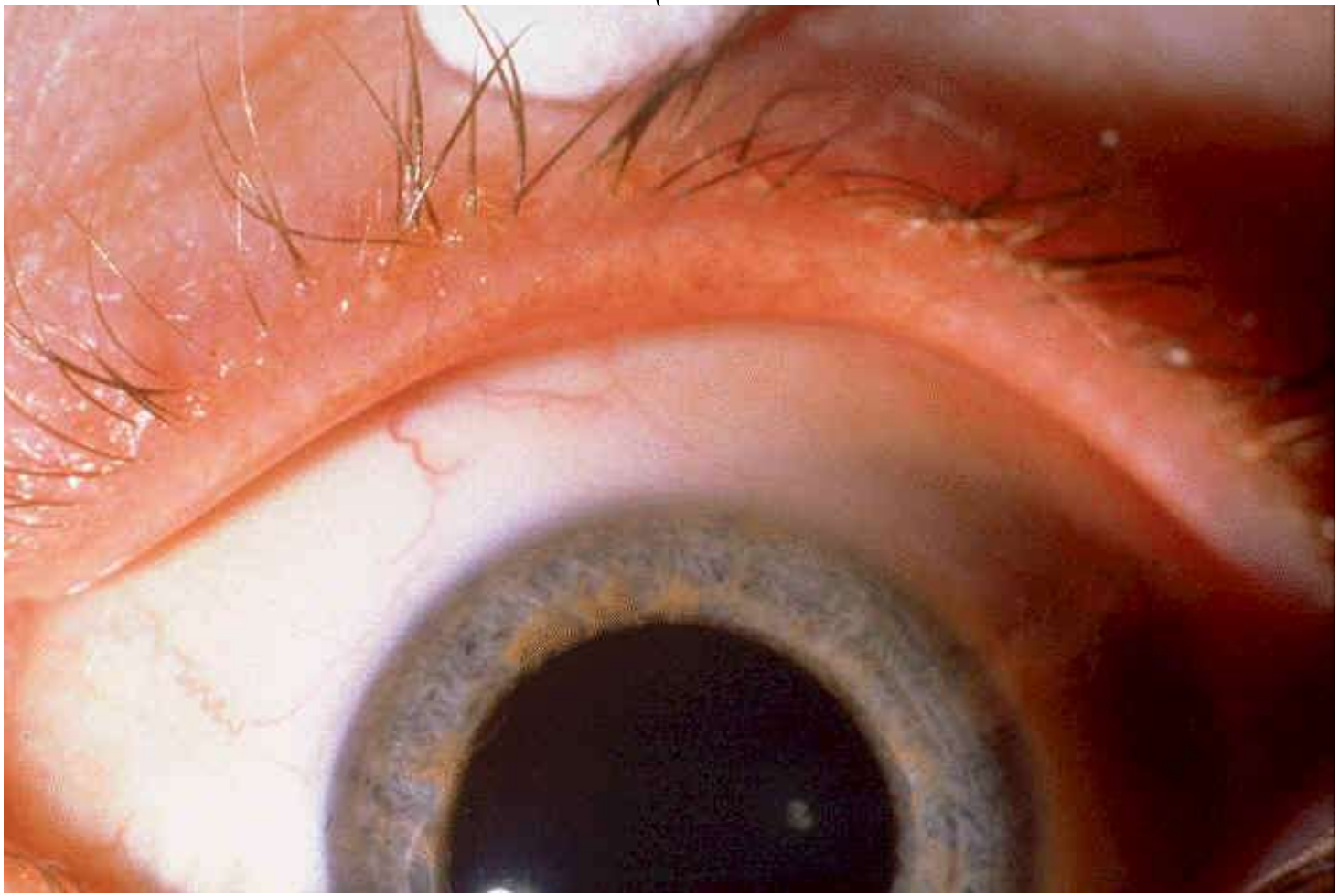
Staphylococcal Blepharitis

- **Infected follicles and madarosis (loss of lashes)**



Staphylococcal Blepharitis

- **Infected follicles** trace to 1



Staphylococcal Blepharitis

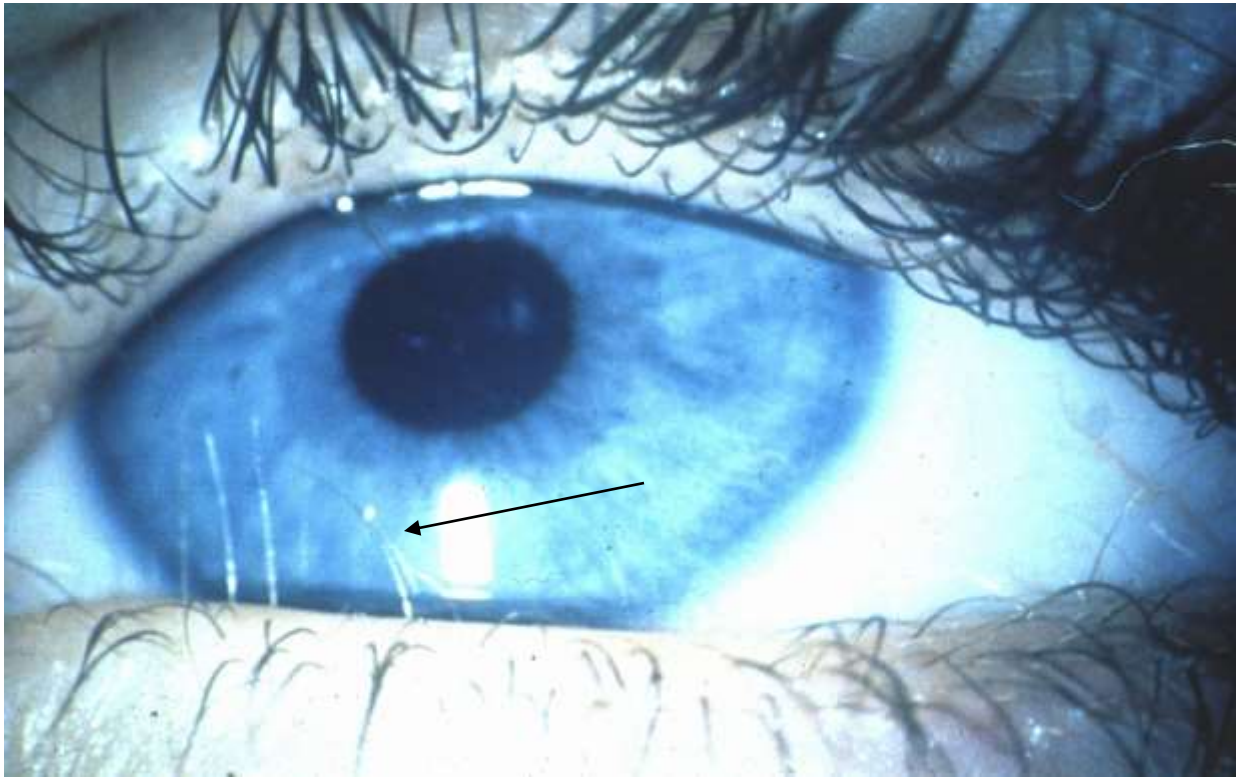
- **Ulcers at the base of the lashes**



15

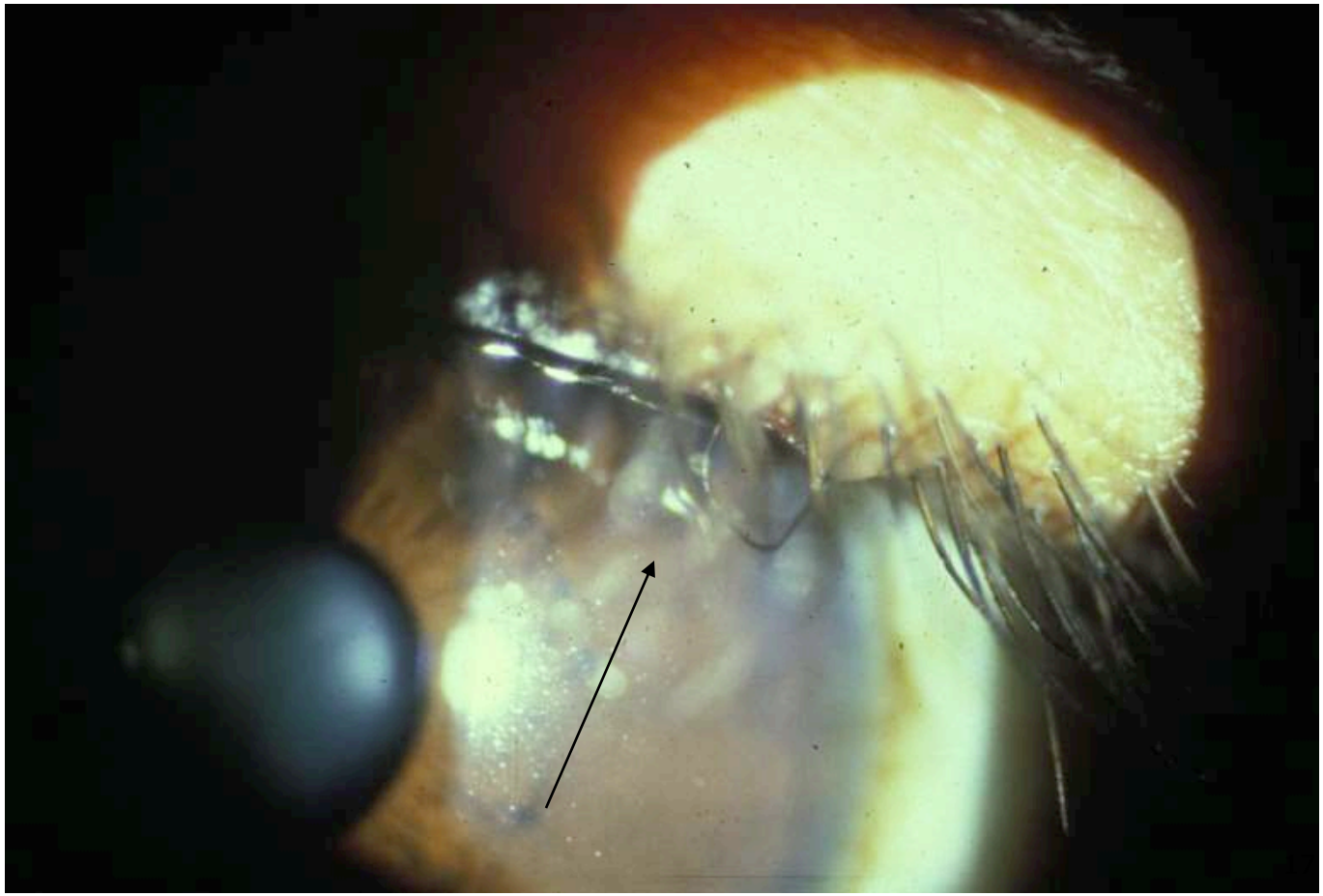
Staphylococcal Blepharitis

- Damage to the lash follicles → trichiasis (lashes that point towards the cornea and may actually scratch the cornea)



Staphylococcal Blepharitis

- **Trichiasis - lashes need to be epilated if touching cornea**



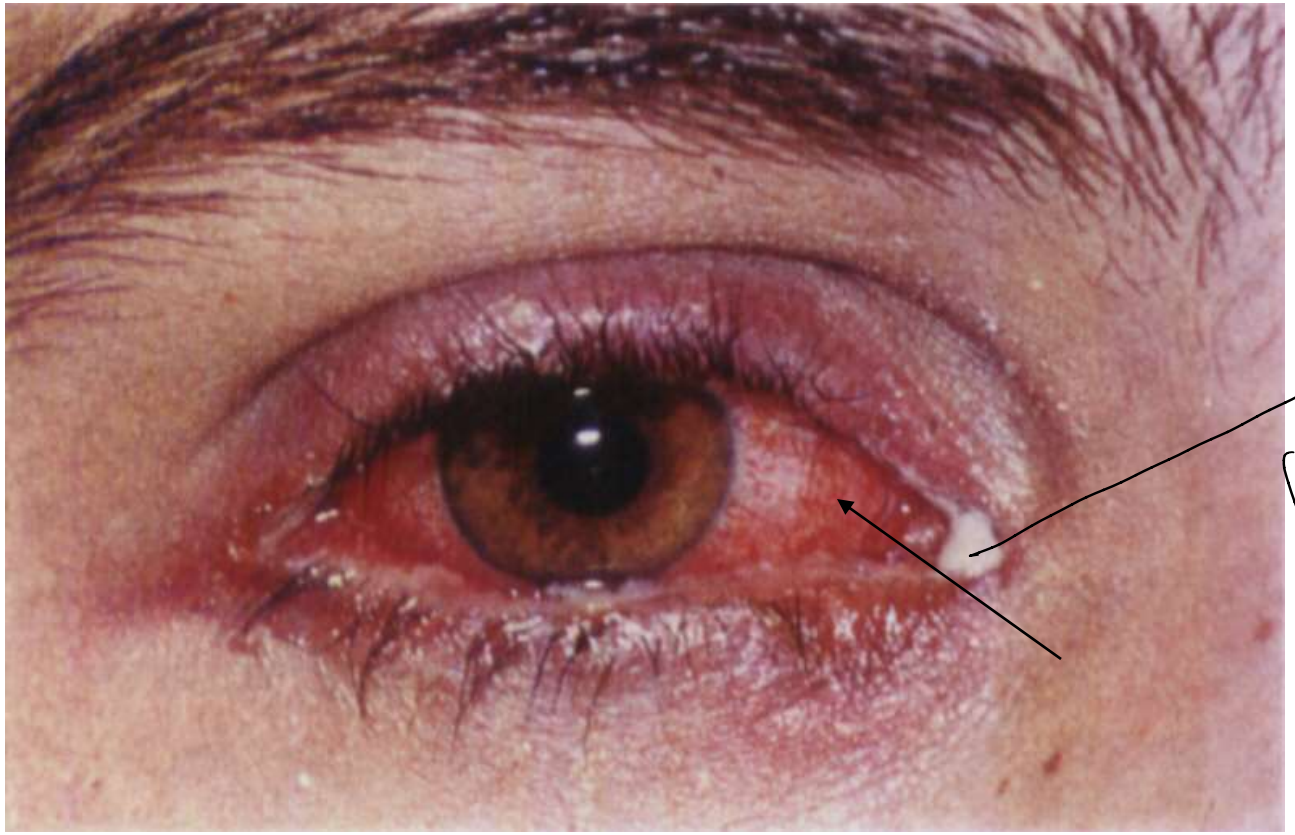
Staphylococcal Blepharitis

- **Red and thickened (tylosis) lid margins**
  - **Seen mostly with chronic disease**
  - **Also note madarosis and conjunctivitis**
- **Lid margins may become indurated (hardened) over time**



Staphylococcal Blepharitis

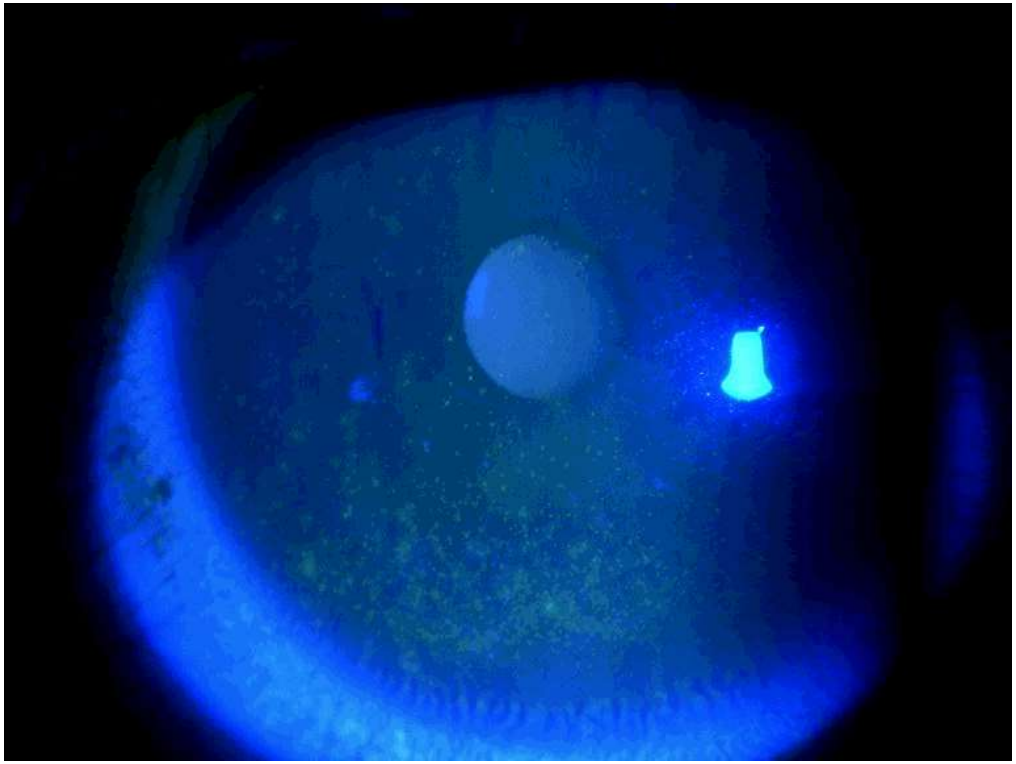
- **May have secondary conjunctivitis (usually papillary)**
  - **Sometimes called staphylococcal blepharoconjunctivitis**



Staphylococcal Blepharitis

(superficial)

- **Punctate epithelial erosions (PEE)**
  - **Result from reaction to bacterial exotoxins**
- **PEE in inferior third of the cornea**
- **PEE stains with fluorescein (green color with colbalt blue filter)**

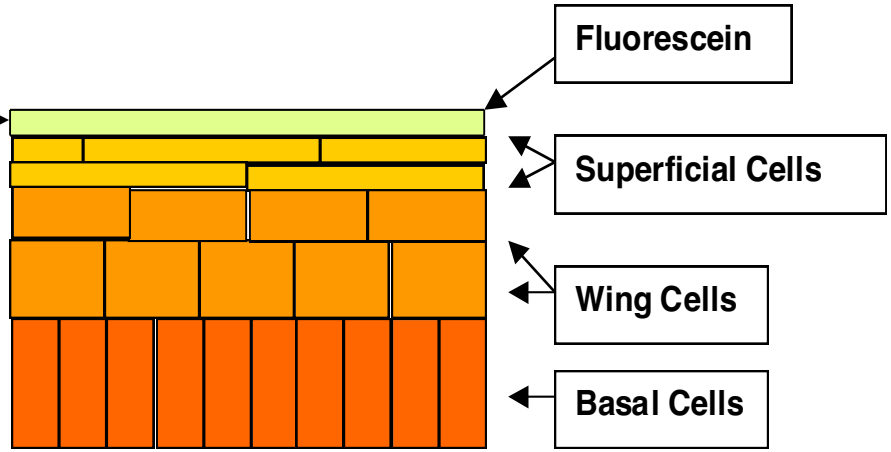


surface damage of cornea.

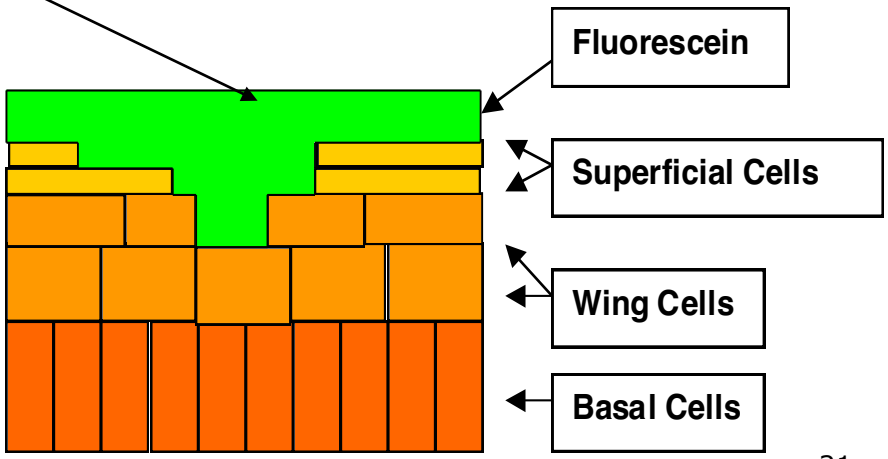
Staphylococcal Blepharitis

EPITHELIUM

- Normal



- PEE (punctate epithelial erosions)
- (AKA SPK - superficial punctate keratopathy)
- Divits in the corneal epithelium that cause fluorescein to fluress



Staphylococcal Blepharitis

- Diagnosis
  - Usually based on clinical findings
  - Severity will dictate appropriate treatment

Staphylococcal Blepharitis

Treatment - lid hygiene

- **Remove crusts** – hot/warm water to moist crusts
- **Remove bacteria** – AMPPP, for sleeping clean bacteria from lid
- **Increase blood circulation to lid margin**
- **Two step process - warm compresses and lid scrubs**



Staphylococcal Blepharitis

Warm compresses

- Soak lids with warm, wet compress (eyelids closed)
- 10-15 min bid-qid – 4x a day
  2x a day

Lid scrubs

- Diluted baby shampoo and water or
- Commercially available cleansers
- Scrub lid margins
- Bid to qid after warm compresses
- Hypochlorous acid 0.02%





Staphylococcal Blepharitis

Baby Shampoo – the 'old' treatment modality

- **The baby shampoo is first diluted one-to-one with water in 'cup' in palm of hand.**
- **Then mixed by rubbing with clean fingertips and applied in gentle oval scrubbing motion to margin and eyelash base of closed eyelid for 1 minute, followed by fresh water facial rinse.**
- **Baby shampoo can alternatively be diluted in container (e.g. plastic cup) and scrubbing performed using washcloth wrapped around finger (after dipping it in the diluted shampoo). A cotton tip applicator may be used alternatively.**
- **NOT recommended anymore due to drying effects**



Hypochlorous acid 0.02%

- Ocusoft HypoChlor™
- Spray or gel
- Used as needed or daily
- Can be used with lid wipes
- Spray form often used AM
- Gel form often used PM
- Generic versions as well

Salt + vinigar + water



Staphylococcal Blepharitis (gels)

Antibiotic ointment applied to lid margins

preservetis kill bacteria

- **Bacitracin**
  - **Cell wall inhibitor**
- **Erythromycin**
  - **Macrolide**
- **Gentamicin**
  - **Aminoglycoside**
- **Tobramycin**
  - **Aminoglycoside**



Regimen

- **Moderate: qhs or bid**
- **Severe: tid or qid**

Prescription Writing

Christina Gosses, OD

354 NW Orange Lane

Chicago, IL 60631

Name Sherman Pierce Age 22 Date 09/22/24

Address 2043 College Way, Forest Grove OR 97116

℞

Bacitracin ophthalmic ung – ointment
Dispense: one 3.5 gram tube
Instructions: apply 1/4 inch to each lid margin of both eyes twice a day for 10 days

X No Refill _____,O.D. _____,O.D.

☐Refill x ___ Dispense as Written Product Selection Permitted

Staphylococcal Blepharitis

Lubricants

- **Dry eye secondary to lid condition**
- **Artificial tears qid or prn**

Oral Antibiotics – usually ROSEISA (rosacea) + Blepharitis

- **Valuable when unusually severe**
- **Doxycycline: 100mg PO bid x 7-10 days**
  - **Tetracyclines inhibit lipase synthesis by bacteria**
  - **Doycycline inhibits Staphylococcal exotoxin-induced cytokines and chemokines**
  - **Are helpful even if bacteria are resistant to antibiotic effect**

Staphylococcal Blepharitis

Follow-up

- **1-3 weeks depending on severity and medications prescribed**
- **Consider follow-up 2-3 weeks later**
- **If cornea involved, follow-up reduced to days** – can loose a sight. To see it is improving.

Patient education

- **Chronic condition - Usually exacerbations and remissions**
- **Goal: to control not cure** – NO CURE!
- **Importance of daily lid hygiene**
- **Medication for acute flare-ups**

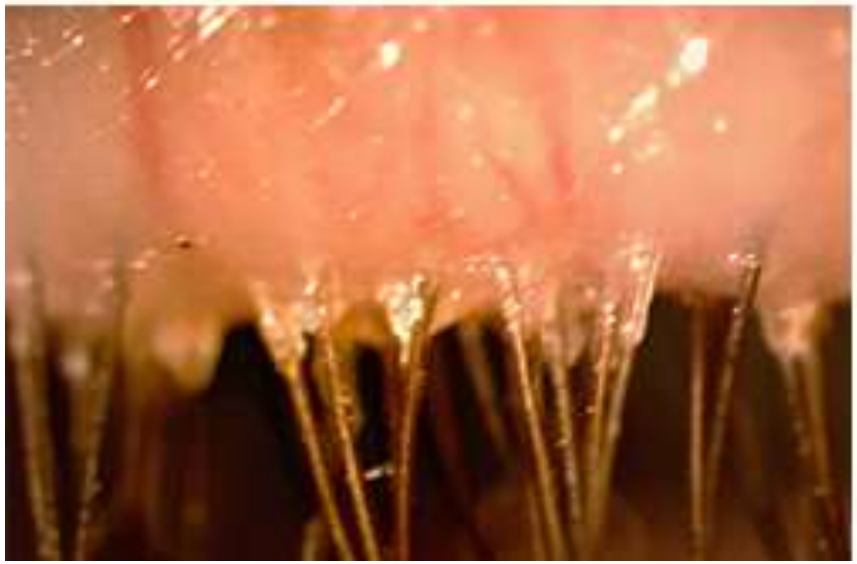
BlephEx – Blepharitis treatment option

- Rysurg
- Medical grade disposable micro-sponge spins along edge of eyelids and lashes
- Removes scurf/debris and exfoliates eyelids
- Proparacaine usually instilled prior to treatment
- Eyes rinsed afterwards
- 6 -8 minute procedure

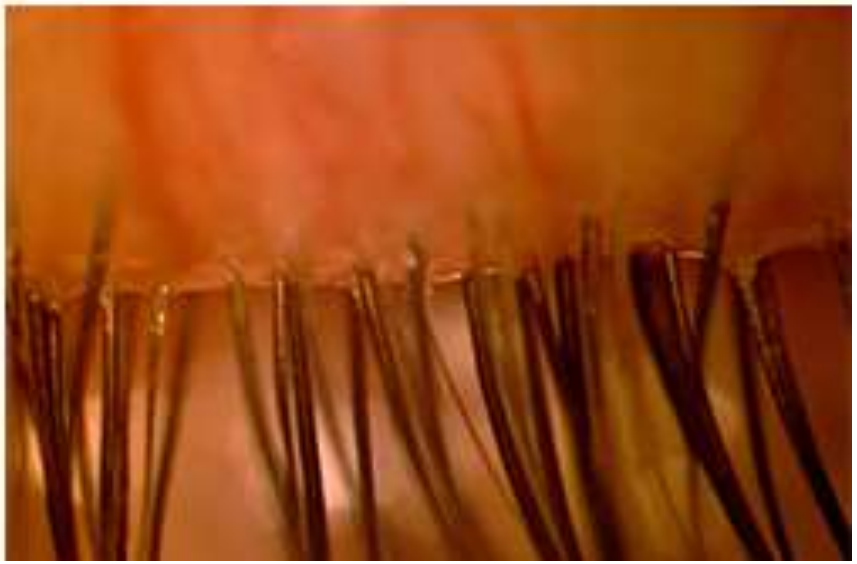


Blephex

Before

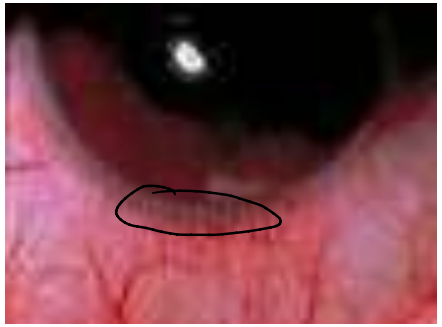


After 8 min of treatment



Staphylococcal Blepharitis Associated Marginal Infiltrate

- White blood cells infiltrate cornea between epithelium and stroma (subepithelial)
  - White spot in corneal periphery with clear area between it and limbus
  - Does not stain with fluorescein
- When the white cells die, they release enzymes that can damage the epithelium above the infiltrate
  - The infiltrate may then stain in a sporadic punctate pattern



Staphylococcal Blepharitis Associated Marginal Infiltrate

Ls further at - limbus

Hypersensitivity to staphylococcal toxins

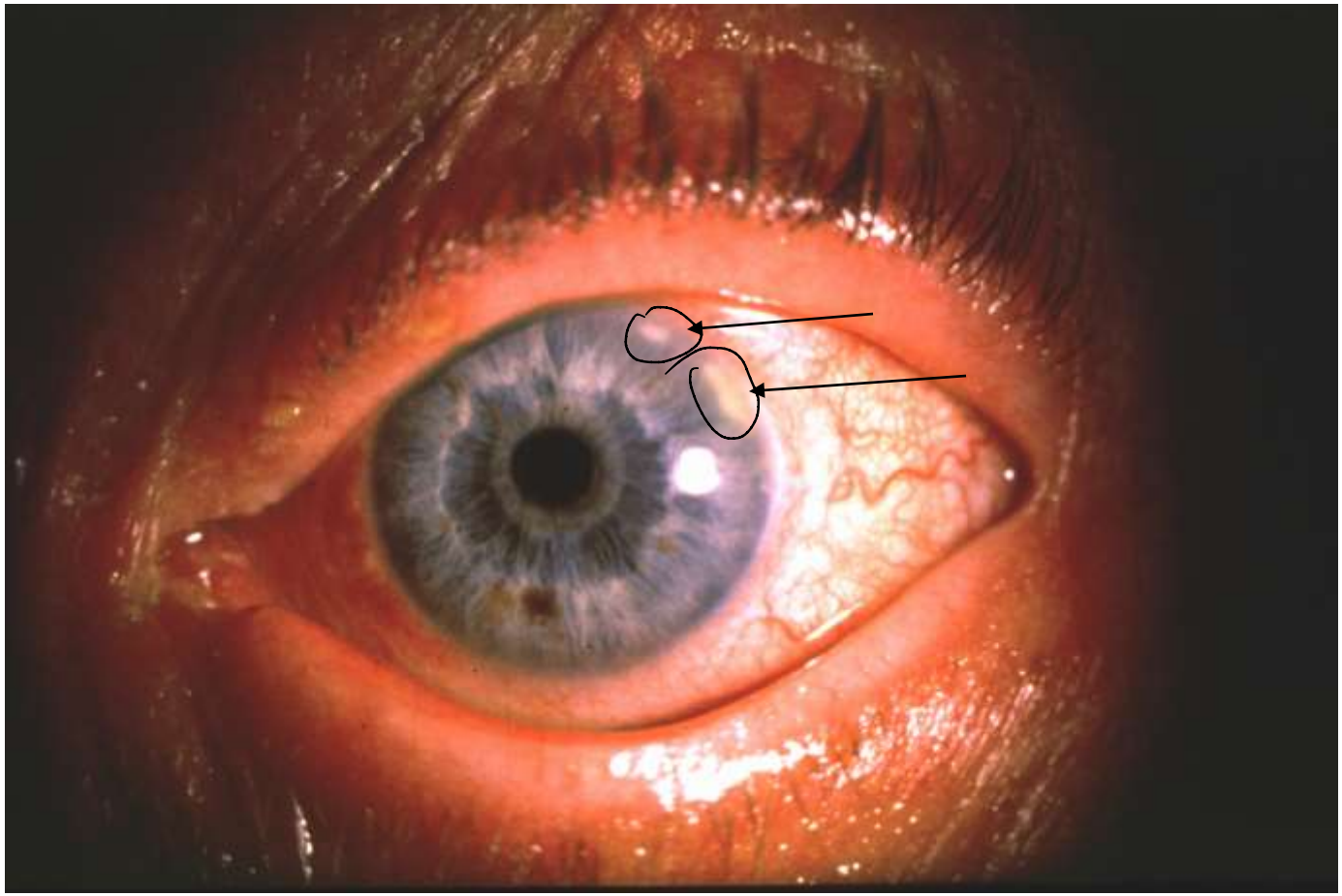
Ag-Ab-Complement triggered

- **Ag (staph product) +**
- **Ab** (ANTI BODIES) **(produced in conjunctiva) +**
- **complement components (from limbal blood vessels) →**
- **C3a and C5a (chemotactic factors for white cells from the limbal blood vessels) →**
- **White spot in cornea of an accumulation of white cells**

Staphylococcal Blepharitis Associated Marginal Infiltrate

- **Marginal infiltrates**

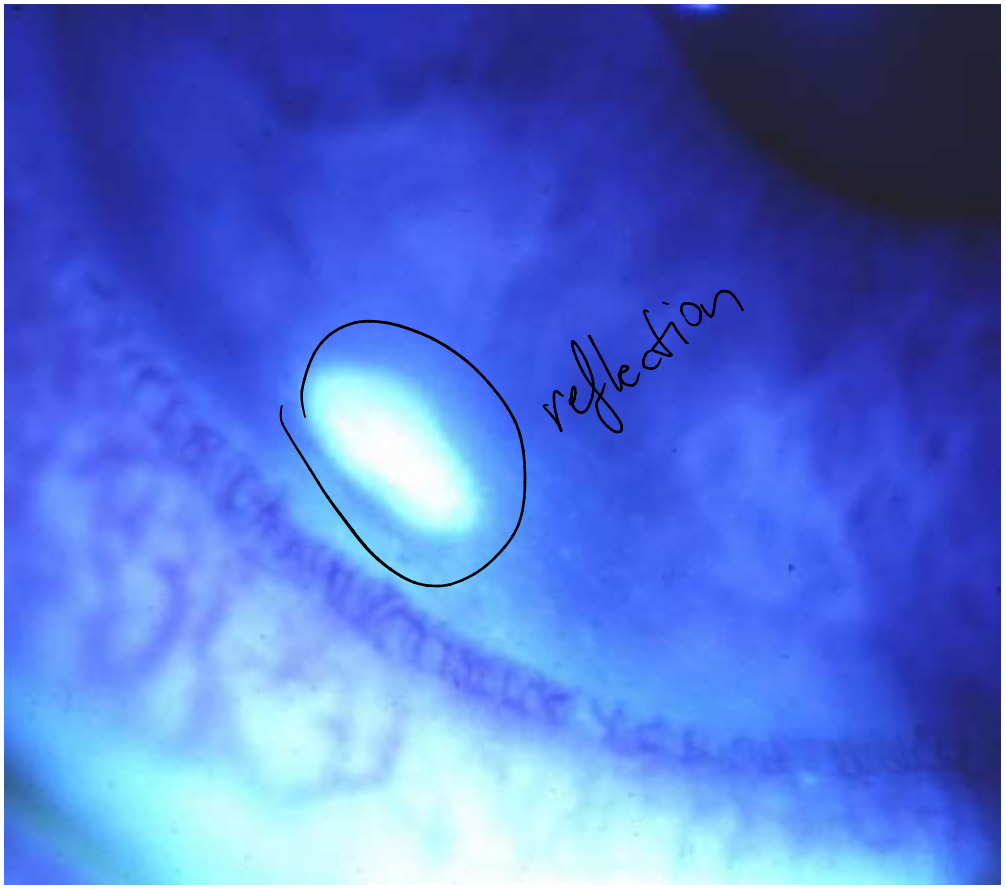
May or may not be seen with fluorescein



Staphylococcal Blepharitis Associated Marginal Infiltrate

A marginal infiltrate that does not stain





Staphylococcal Blepharitis Associated Marginal Infiltrate

Treatment

- **Treat the staphylococcal blepharitis and add:**

- **Topical steroid q2h to qid**
  - **Prednisolone acetate**
  - **Dexamethasone**
  - **Loteprednol**

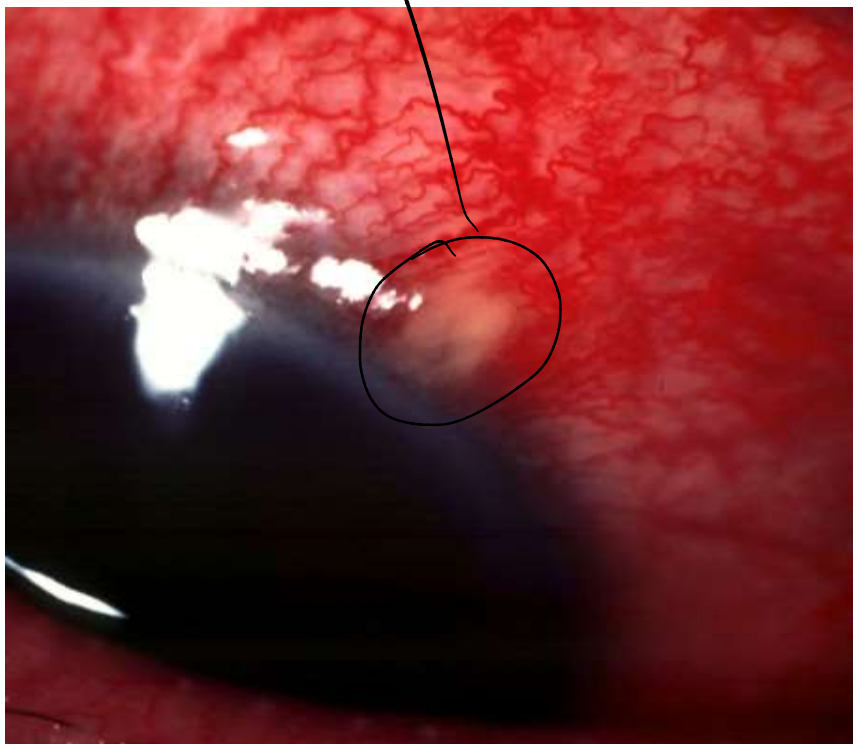
Or…



- **Topical antibiotic/steroid combination such as tobramycin with dexamethasone (Tobradex), or tobramycin with loteprednol (Zylet) q2h to qid**

Staphylococcal Blepharitis Associated Phlyctenule

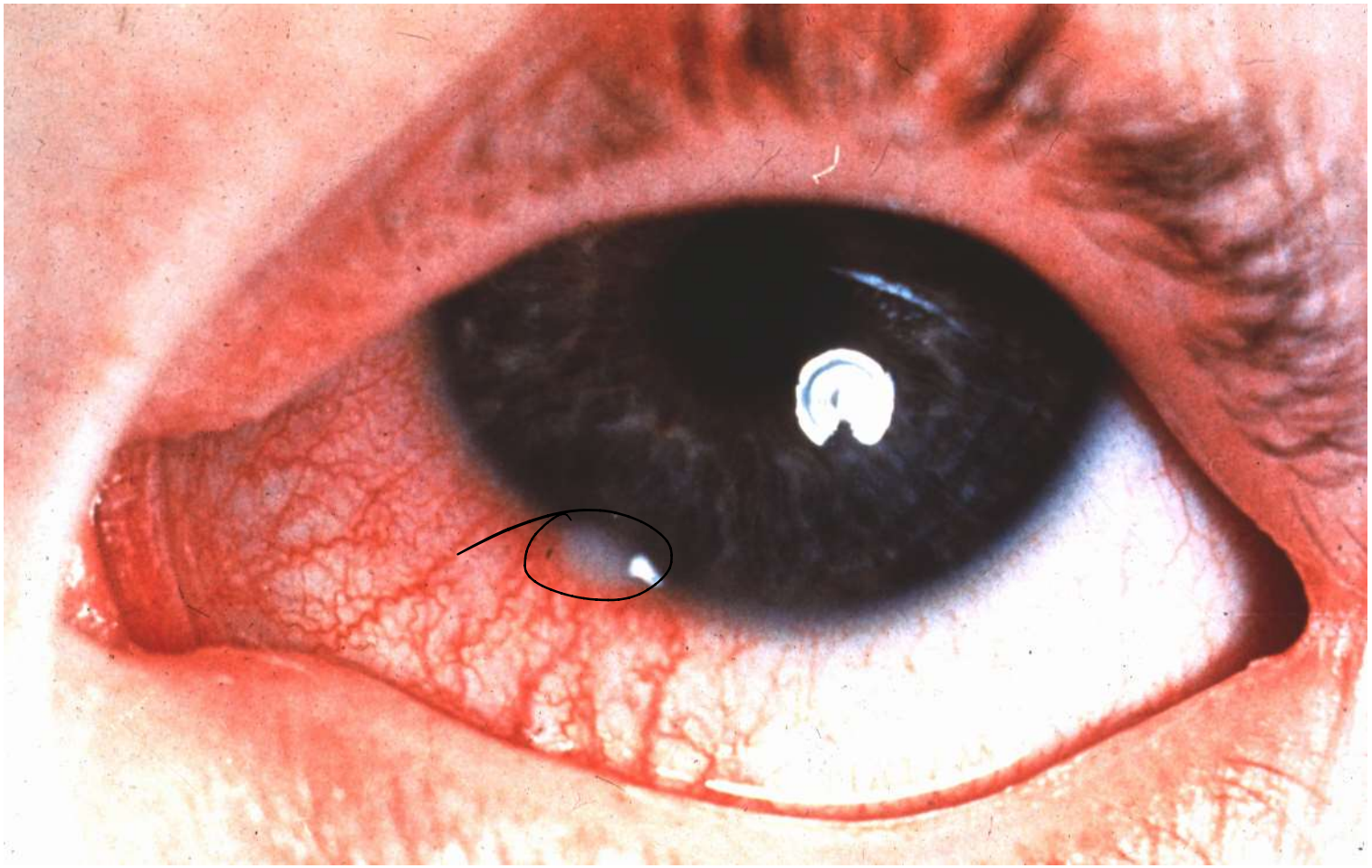
- Hypersensitivity reaction to staph toxins
- Type IV reaction
- Raised white lesion on conjunctiva, limbus, or cornea
- Can move over cornea, pulling blood vessels and scarring as it goes



because it is on the limbus

STEROID

Staphylococcal Blepharitis Associated Phlyctenule



Not quit in the cornea

Staph Blepharitis Associated Phlyctenule

Treatment

- **Treat the staphylococcal blepharitis and add:**

- **Topical steroid q2h to qid**
  - **Prednisolone acetate**
  - **Dexamethasone**
  - **Loteprednol**



Or…

- **Topical antibiotic/steroid combination such as tobramycin with dexamethasone, or tobramycin with loteprednol q2h to qid**

Staph Blepharitis Associated Phlyctenule

- Note – several years ago a primary cause of phlyctenule was tuberculosis (TB)
- Patients who were:
  - Recent immigrants from areas endemic with TB, or
  - Individuals who acquired active TB locally
- Now, Staphyloccus has once again become a primary cause of phlyctenule

Tuberculosa

Seborrheic Blepharitis

autoimune reaction similar to staf.

Etiology

- Unknown
- Often associated with seborrheic dermatitis
  - Chronic disorder of the skin/head areas where sebaceous glands are abundant
  - Dandruff (AKA scurf) - excessive shedding of dead skin cells of the scalp - is often a symptom of seborrheic dermatitis



Seborrheic Blepharitis

Subjective

- **Frequently asymptomatic**
- **Possible morning “mattering”**
- **Dry eye complaints (poor tear film)**
- **Symptoms may depend on other concurrent processes**

Seborrheic Blepharitis

Objective

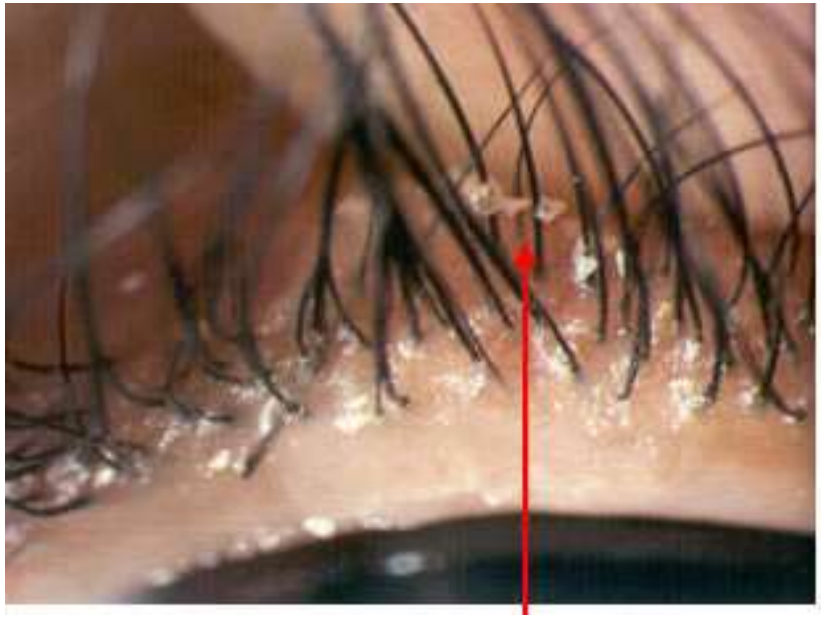
- **Greasy, waxy scales at the base of the lashes that flake off easily (dandruff-like)(AKA scurf)**
- **Skin of the lid a little greasy**



Seborrheic Blepharitis



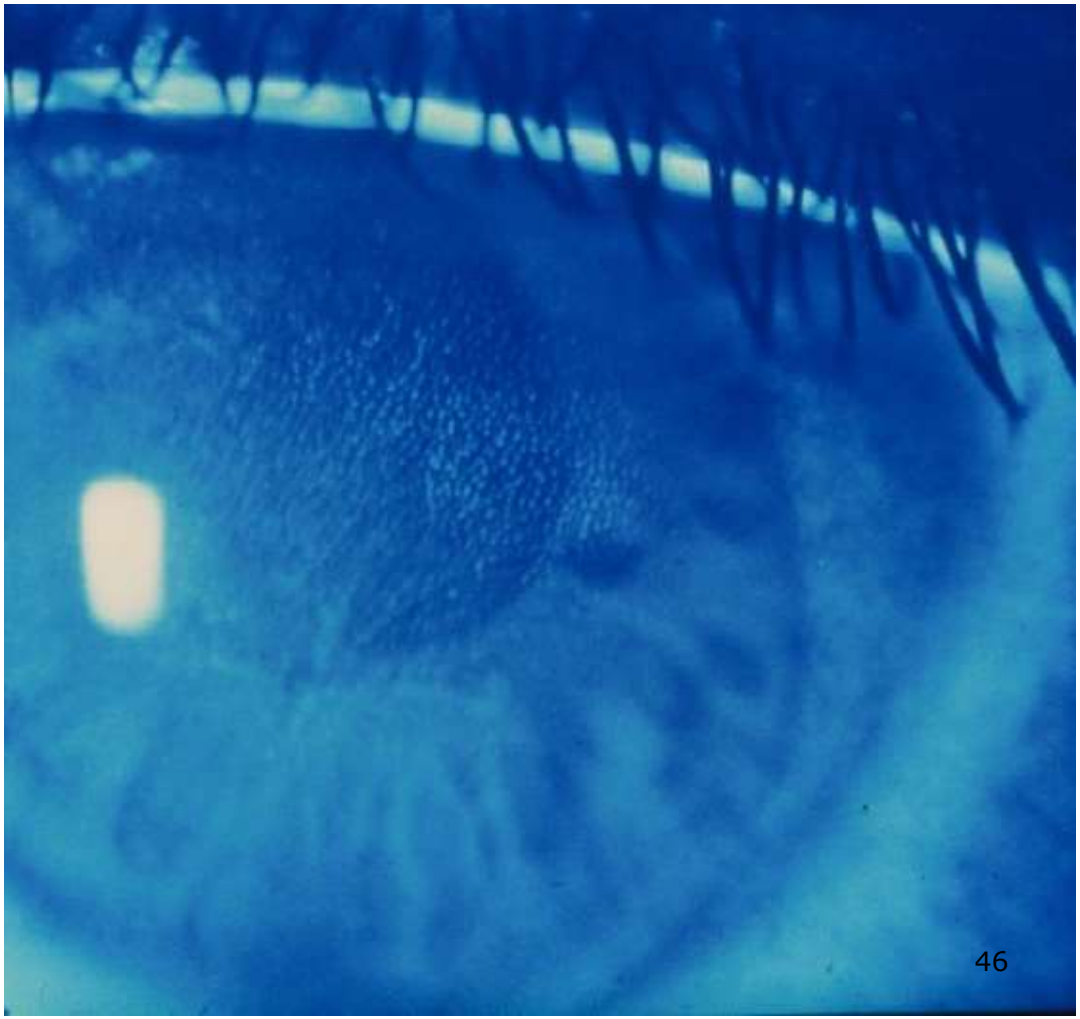
More wet



Seborrheic Blepharitis

- **Interpalpebral PEE**

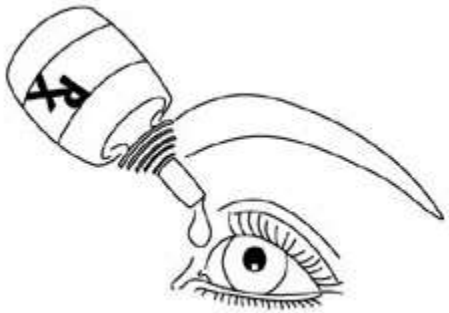
more in the middle of cornea



Seborrheic Blepharitis

Treatment:

- **Lid hygiene**
- **Lubricants**
- **Follow-up 2-4 weeks**
- **Patient education on chronicity**
  - **Lid hygiene daily**





Mixed Blepharitis

- Seborrheic blepharitis with associated staphylococcal superinfection
- Plan
  - Same as for staph blepharitis

Meibomian Seborrheic Blepharitis

Etiology

- **Unknown**
- **Excessive oil production**
- **Quality of oil is abnormal**

Subjective

- **Dry eye complaints**

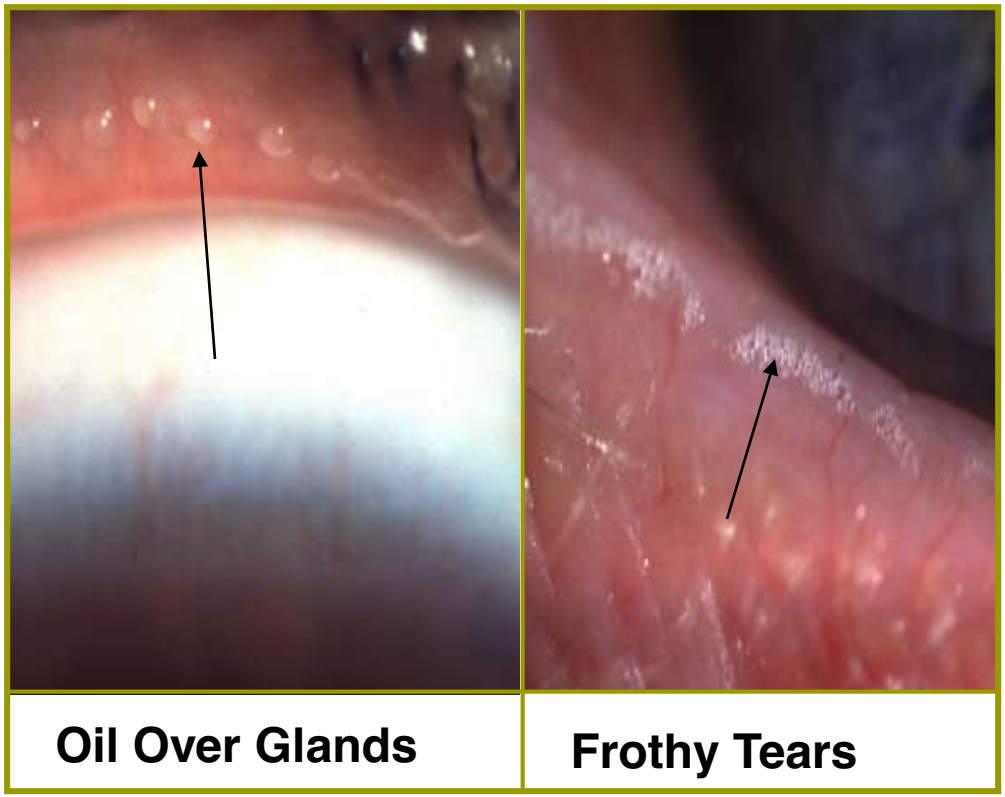
Meibomian Seborrheic Blepharitis

Objective

- Frothing of tears
- "Oil slick" in tear film
- TBUT may be reduced
- Conjunctival injection
- Lid tissue appears normal

Treatment

- Keep lid margins clean
- Lubricants



Oil Over Glands | Frothy Tears

bubbly

severe treatment

Angular Blepharitis

- **Localized eczematoid inflammation of lid at outer canthus and sometimes medial canthal region**



on lateral site

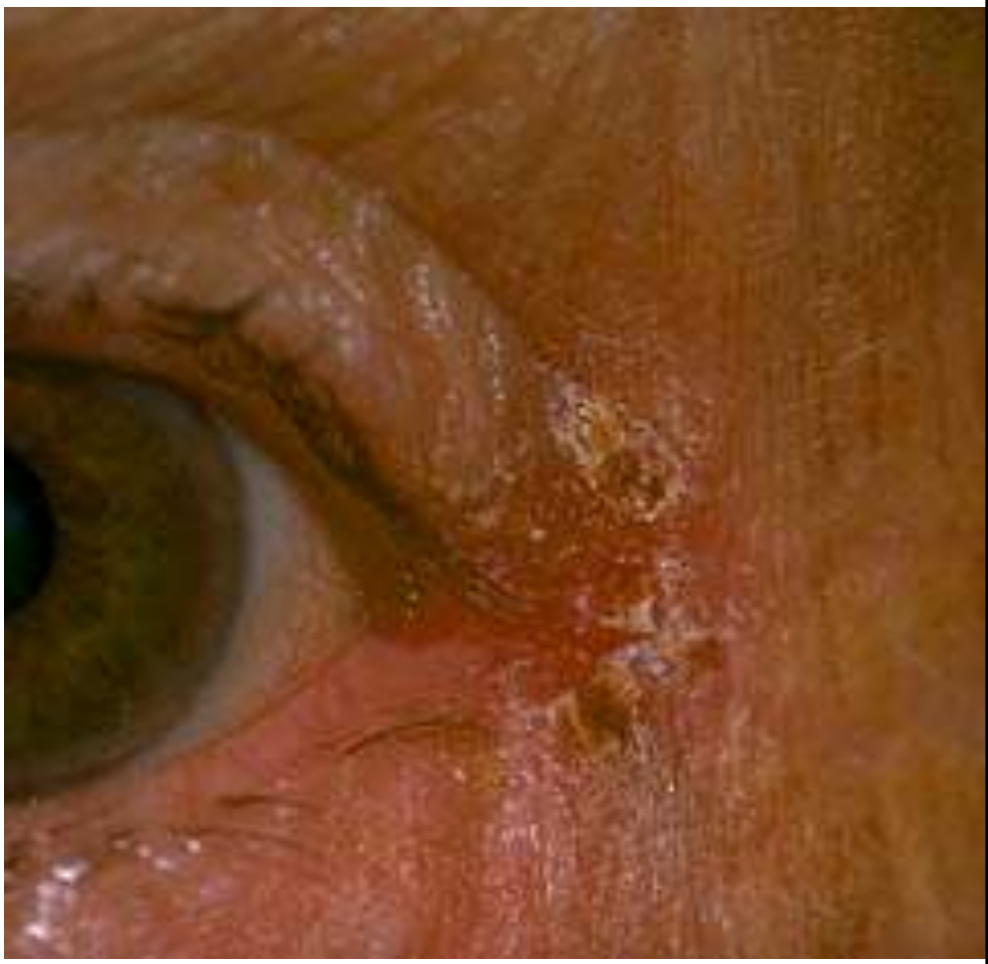
- **Staphylococcal form:**
  - **Dry and scaly**
- **Moraxella form:**
  - **Wet**
  - **Macerated lid with whitish, frothy discharge**

some different bacteria involved

- **Angular blepharitis may involve conjunctiva**
- **All forms of angular blepharitis call for treatment with antibiotic ointment**
  - **May need to try different ABs based on bacteria type**

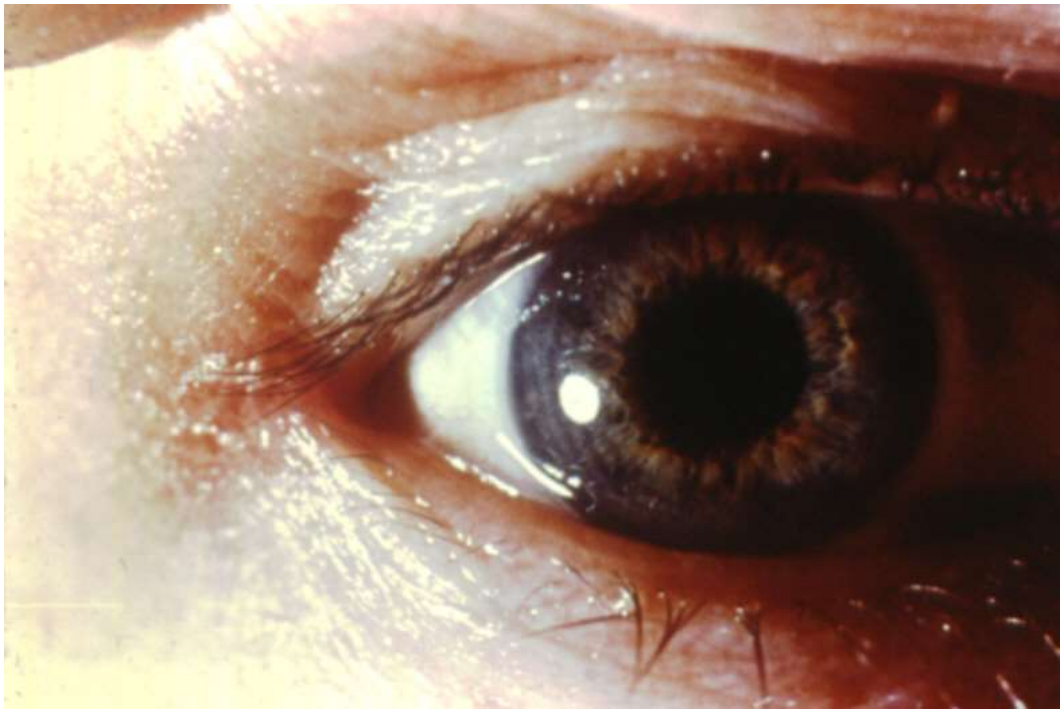
Angular Blepharitis

- Chronic lid irritation
- Skin drying, scaling, cracking in outer canthi
- Cause
  - *Moraxella lacunata*
  - *Staphylococcus epidermidis*
  - *Staphylococcus aureus*



Angular Blepharitis

- ***Moraxella*: zinc sulfate 0.25% solution**
- ***Staphylococcus*: bacitracin or erythromycin**



THANK YOU!